Analysing Smart City Bike Sharing Data using Power BI

1. Project Overview

This project involves cleaning, transforming, and analysing raw data and creating an interactive **Power BI dashboard** to derive meaningful business insights.

Objective

The main objective is to demonstrate data pre-processing techniques and an interactive Power BI dashboard visualization to make informed decisions.

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2. Problem Statement

- Determine which cities have the highest **System Imbalance Scores**, indicating poor bike distribution efficiency.
- Pinpoint geographic clusters where stations are consistently **"Critical: Empty"** or **"Critical: Full,"** preventing users from renting or returning bikes.
- Identify "outlier" stations where the number of stands significantly exceeds the number of available bikes, leading to **wasted infrastructure**.
- Use data to prioritize which cities require manual bike redistribution to move the network from a "Critical" state back to a **"Balanced"** state.

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4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Station Number	Whole Number	Number for each Station
Station	Text	Name of each Station
Station Address	Text	Address of the Station
Latitude	Decimal Number	Latitude points of the location
Longitude	Decimal Number	Longitude points of the location
Banking	Text	On-site credit card payments
Bonus	Text	"Bonus" time or credits to users
Status	Text	Opened or closed status of the Station
City	Text	Station City
Bike Stands	Whole number	No. of bike stands in the Station
Available Bike Stands	Whole Number	No. of bike stands ready to park
Available bikes	Whole number	No. of bikes available to rent

Last update Date and Time	Date, Time	Records the exact date and time the station's server last communicated its "Available Bikes" and "Stands" count to the central system.
Station Status	Text	Tells you whether a bike station is currently available for public use or restricted due to maintenance or system issues.
Station Size	Text	Groups stations based on their total physical capacity (Total Bike Stands).

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5. Tools & Technologies

- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

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6. Data Pre-Processing (Power Query)

Tasks Performed:

- **Data Cleaning & Transformation:** Handled missing values, standardized formats, and created calculated fields.
- **Filtering & Sorting:** Organized data to focus on relevant records.
- Convert the data into Fact and Dimension Table

❖ Separated the Name column with number to unique texts

	Number	Name	Addresss	Position
1	2000	TEST DSI PLAISIR	null	46.5488766782771,
2	1297	ST FÉRRÉOL DAVSO	ST FERREOL DAVSO - RUE FRANCIS DAVSO ANGLE RUE SAINT FERREOL	43.2934857361865
3	8265	PAULET	NEGRESKO PAULET - FACE AU 42 RUE NEGRESKO	43.2693185563711
4	8065	PROMENADE POMPIDOU PA...	PROMENADE POMPIDOU PALM BEACH - 6 PROMENADE GEORGES PO...	43.2672725156718
5	1011	BORNE TEST NANTES 1	Borne test 1	47.19504, -1.5575
6	4342	FLAMMARION MONTRICHET	FLAMMARION MONTRICHET - PLACE LE VERRIER ANGLE BD FLAMMAR...	43.3059934537668
7	6176	CANTINI ROUET	CANTINI ROUET - FACE AU 27 AVENUE JULES CANTINI	43.2847759836122
8	3322	ARENC CHANTERAC	32 rue de Chanterac 13003 Marseille	43.3123370339008,
9	1264	CANEBIÈRE DUGOMMIER	CANEBIERE DUGOMMIER - LA CANEBIERE ANGLE BOULEVARD DUGO...	43.2978963132934
10	1191	CANEBIEREST FERREOL	CANEBIERE SAINT FERREOL - ANGLE CANEBIERE SAINT FERREOL	43.2962076706139
11	5058	SAINT PIERRE	SAKAKINI SAINT PIERRE - BOULEVARD SAKAKINI ANGLE RUE SAINT PIE...	43.2928008635458
12	4293	BLANCARDE	BLANCARDE - GARE DE LA BLANCARDE	43.2962634051705
13	1245	LAFAYETTE	GAMBETTA LAFAYETTE - 6 ALLEES LEON GAMBETTA	43.2990279382647
14	7153	CORSE ST MAURICE	CORSE SAINT MAURICE - 28 AVENUE DE LA CORSE ANGLE RAMPE SAIN...	43.2894676088998
15	6154	PIERRE PUGET BRETEUIL	PIERRE PUGET BRETEUIL - COURS PIERRE PUGET ANGLE RUE BRETEUIL	43.2903297702063
16	1002	HOTEL DE VILLE	HOTEL DE VILLE - Face au 42 QUAI DU PORT	43.2961387765145,
17	1224	BIBLIOTHEQUE	PLAINE BIBLIOTHEQUE - 45 RUE DE LA BIBLIOTHEQUE	43.2958251832795
18	1287	RÉFORMÉS	STALINGRAD REFORMES - SQUARE STALINGRAD	43.2995209755801
19	500	IT PLAISIR	null	48.799828, 1.94685
20	5220	ORAN	ROOSEVELT ORAN - 53 COURS FRANKLIN ROOSEVELT ANGLE RUE D'OR...	43.2989919565036

❖ Replaced the null values in address column with the names in Name column

	Number	Name	Address	Position.1
1	2000	TEST DSI PLAISIR	TEST DSI PLAISIR	46.5488766782771
2	1297	ST FÉRRÉOL DAVSO	ST FERREOL DAVSO - RUE FRANCIS DAVSO ANGLE RUE SAINT FERREOL	43.2934857361865
3	8265	PAULET	NEGRESKO PAULET - FACE AU 42 RUE NEGRESKO	43.2693185563711
4	8065	PROMENADE POMPIDOU PA...	PROMENADE POMPIDOU PALM BEACH - 6 PROMENADE GEORGES PO...	43.2672725156718
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20	5220	ORAN	ROOSEVELT ORAN - 53 COURS FRANKLIN ROOSEVELT ANGLE RUE D'OR...	43.2989919565036

❖ Separated Latitude and Longitude column using Delimiter.

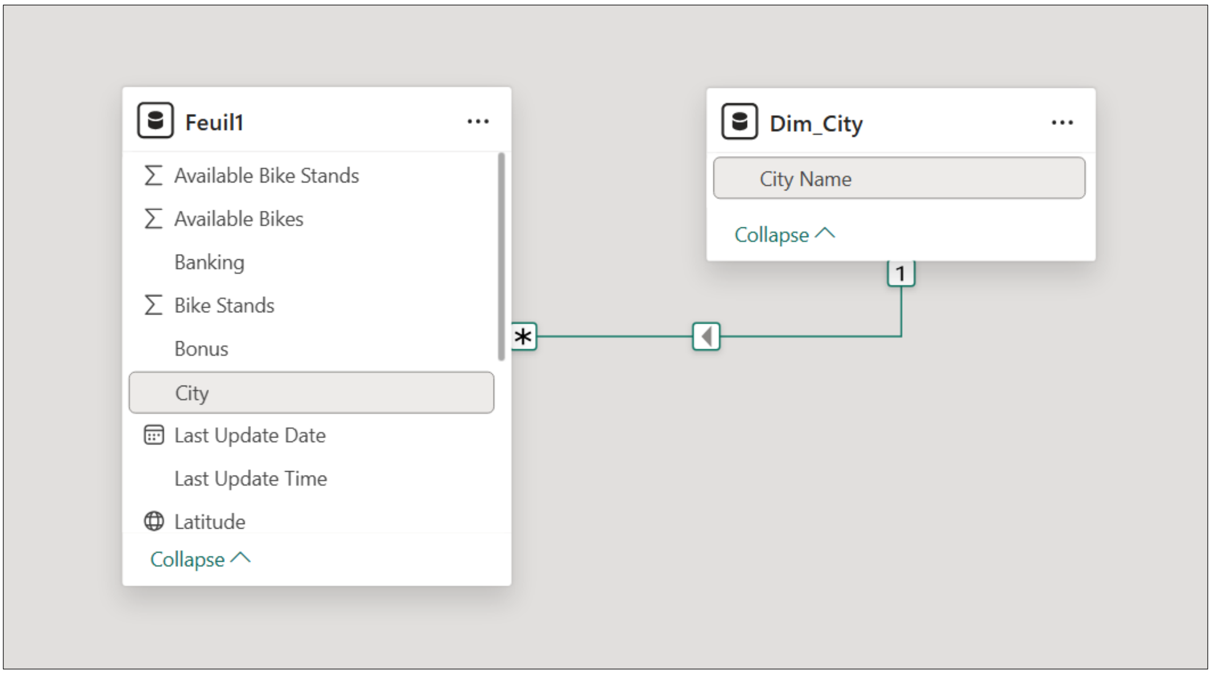
	Latitude	Longitude	Banking	Bonus	
32	50.813938	4.330966	False	True	OP
an 1	50.812325	4.427962	False	False	OP
	53.3488	-6.281637	False	False	OP
	39.45027304	-0.33336296	False	False	OP
	39.49801319	-0.377294117	False	False	OP
	43.623823	1.475935	True	False	OP
	39.4704341	-0.374915095	False	False	OP
	50.460596	4.842836	False	False	OP
	53.355954	-6.278378	False	False	OP
	45.763244	4.778641	False	False	OP
NFEESTLAAN	50.895485	4.341311	False	False	OP
	50.834237	4.350732	False	False	OP
	43.615988	1.42163	False	False	OP
	39.48153014	-0.352239033	False	False	OP
	43.590111	1.423003	False	False	OP
	53.340714	-6.308191	False	False	OP
	45.805207	4.88477	False	False	OP
	39.45166003	-0.388669126	False	False	OP
	43.611556	1.439662	False	False	OP
	39.49294616	-0.401981188	False	False	OP

❖ Separated Date and Time using Delimiter

	e Stands	Available Bike Stands	Available Bikes	Last Update Date	Last Update Time
1	20	15	2	12-11-2025	14:48:26
2	30	23	7	12-11-2025	14:49:46
3	30	30	0	12-11-2025	14:55:01
4	15	0	15	12-11-2025	14:47:54
5	15	14	1	12-11-2025	14:47:59
6	20	10	10	12-11-2025	14:53:52
7	26	11	15	12-11-2025	14:47:58
8	15	7	8	12-11-2025	14:56:08
9	36	36	0	12-11-2025	14:52:29
10	12	12	0	12-11-2025	14:49:54
11	25	21	4	12-11-2025	14:48:33
12	25	12	12	12-11-2025	14:49:41
13	22	16	4	12-11-2025	14:52:26
14	16	15	1	12-11-2025	14:48:03
15	13	5	7	12-11-2025	14:54:10
16	40	22	18	12-11-2025	14:50:45
17	12	3	9	12-11-2025	14:55:12
18	20	11	9	12-11-2025	14:48:02
19	21	19	2	12-11-2025	14:52:58
20	15	10	5	12-11-2025	14:48:07

7. Data Modelling

- **Data Model:** Established relationships between tables and defined cardinality.



- **Calculated Columns:**
  - ❖ Created Station Status column to check whether a bike station is currently available for public use or restricted due to maintenance or system issues.

```
Station_Status = switch(true(),
Feuil1[Available Bikes]=0,"Critical: Empty", Feuil1[Available Bike Stands]=0, "Critical:Full", Feuil1[Available Bikes] <=2,"Low Stock","Healthy")
```

Station Address	Latitude	Longitude	Last Update Date	Last Update Time	Station_Status	Station_Size
ANDE - Aprox. C/ San Salvador	37.375915	-5.979015	12 November 2025	14:56:56	Healthy	Medium
ıina C/ Calle Ramón Areces	37.384221	-5.945175	12 November 2025	14:56:19	Low Stock	Medium
ɔx. C/ Palos de la Frontera	37.3811984466081	-5.99352820936488	12 November 2025	14:57:37	Healthy	Medium
Aprox. Avda Carlos V	37.3797214442824	-5.9838011743874	12 November 2025	14:49:17	Healthy	Medium
CANO - Aprox. C/ Blas Infante	37.374136	-6.008866	12 November 2025	14:47:55	Healthy	Medium
QUINA C/ MIGUEL JIMENEZ HINOJOSA	37.3654957	-5.964101	12 November 2025	14:47:51	Healthy	Medium
- Aprox. C/ Cardenal Ilundaín	37.3647503853737	-5.98145215748501	12 November 2025	14:48:22	Critical:Full	Medium
Aprox. C/ Bailén	37.3904564818945	-5.99810323075327	12 November 2025	14:49:17	Low Stock	Medium
Aprox. Nuestra Señora de los Dolores	[illegible]	[illegible]9608370488	12 November 2025	14:56:12	Healthy	Medium
CITY - Aprox. Avda El Greco	37.393129	-5.970594	12 November 2025	14:56:26	Healthy	Medium
ROT - Aprox. Hospital Universitario Virgen del Rocio	37.3614303720606	-5.98160315609194	12 November 2025	14:56:36	Healthy	Medium
ROS - Aprox. C/ Mediterráneo	37.419198	-5.963082	12 November 2025	14:48:49	Critical:Full	Medium
L BUENO MONREAL - Aprox. Avda Manuel Siurot	37.3680453972776	-5.98485417132949	12 November 2025	14:56:44	Healthy	Medium
ɔrox. C/ Aorno	37.398601	-5.973632	12 November 2025	14:55:56	Low Stock	Medium
- Aprox. C/ Luis Rey Romero	37.3955195005648	-6.0023212484675	12 November 2025	14:55:33	Low Stock	Medium
IO DE PADUA - Aprox. C/ Marqués de la Mina	37.3978400223044	-5.99850468687796	12 November 2025	14:56:07	Healthy	Medium
AL COLÓN - Aprox. Torre del Oro	37.381535447211	-5.99555221666194	12 November 2025	14:54:55	Critical:Full	Medium
SPUCIO - Aprox. Calle de Jacques Cousteau	37.407524	-6.007428	12 November 2025	14:54:38	Healthy	Medium

CALLE SAN PABLO - Aprox. C/ Bailén

❖ Created Station size column to group stations based on their total physical capacity (Total Bike Stands).

```
Station_Size = if(Feuil1[Bike Stands] >30, "Large", if(Feuil1[Bike Stands] >15, "Medium", "Small"))
```

Station Address	Latitude	Longitude	Last Update Date	Last Update Time	Station_Status	Station_Size
ANDE - Aprox. C/ San Salvador	37.375915	-5.979015	12 November 2025	14:56:56	Healthy	Medium
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- Aprox. C/ Cardenal Ilundaín	37.3647503853737	-5.98145215748501	12 November 2025	14:48:22	Critical:Full	Medium
Aprox. C/ Bailén	37.3904564818945	-5.99810323075327	12 November 2025	14:49:17	Low Stock	Medium
Aprox. Nuestra Señora de los Dolores	37.3700534164291	-5.95949608370488	12 November 2025	14:56:12	Healthy	Medium
CITY - Aprox. Avda El Greco	37.393129	-5.970594	12 November 2025	14:56:26	Healthy	Medium
ROT - Aprox. Hospital Universitario Virgen del Rocio	37.3614303720606	-5.98160315609194	12 November 2025	14:56:36	Healthy	Medium
ROS - Aprox. C/ Mediterráneo	37.419198	-5.963082	12 November 2025	14:48:49	Critical:Full	Medium
L BUENO MONREAL - Aprox. Avda Manuel Siurot	37.3680453972776	-5.98485417132949	12 November 2025	14:56:44	[illegible]	Medium
ɔrox. C/ Aorno	37.398601	-5.973632	12 November 2025	14:55:56	Low Stock	Medium
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IO DE PADUA - Aprox. C/ Marqués de la Mina	37.3978400223044	-5.99850468687796	12 November 2025	14:56:07	Healthy	Medium
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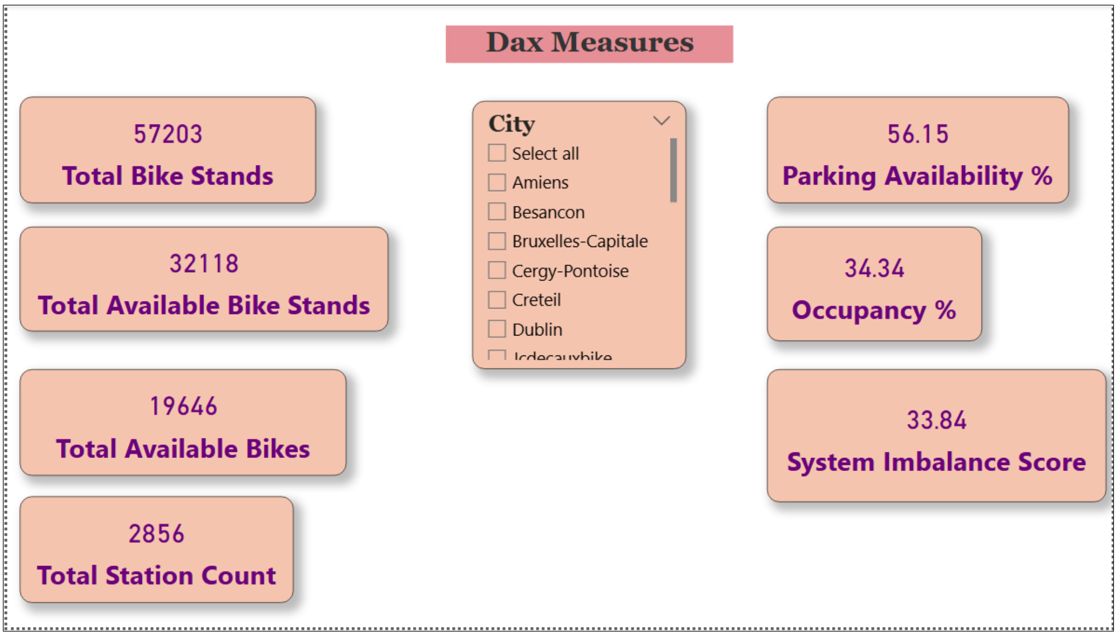
SPUCIO - Aprox. Calle de Jacques Cousteau	37.407524	-6.007428	12 November 2025	14:54:38	Healthy	Medium
DATO - Aprox. C/ Marqués del Nervión	37.3820754594393	-5.96812112073878	12 November 2025	14:55:52	Healthy	Medium
ORTE - Aprox. C/ de Rimas	37.413162	-5.985356	12 November 2025	14:55:54	Critical: Empty	Medium

14:48:49

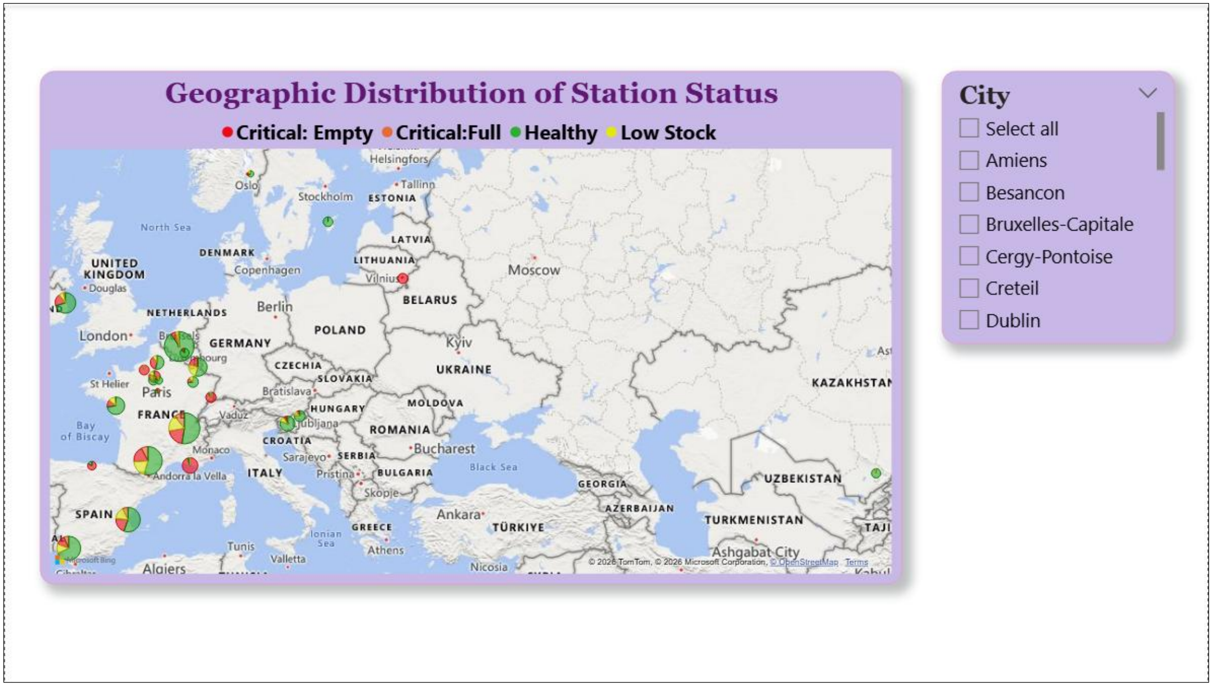
- **DAX Measures:**



8. Analysis and Visualizations

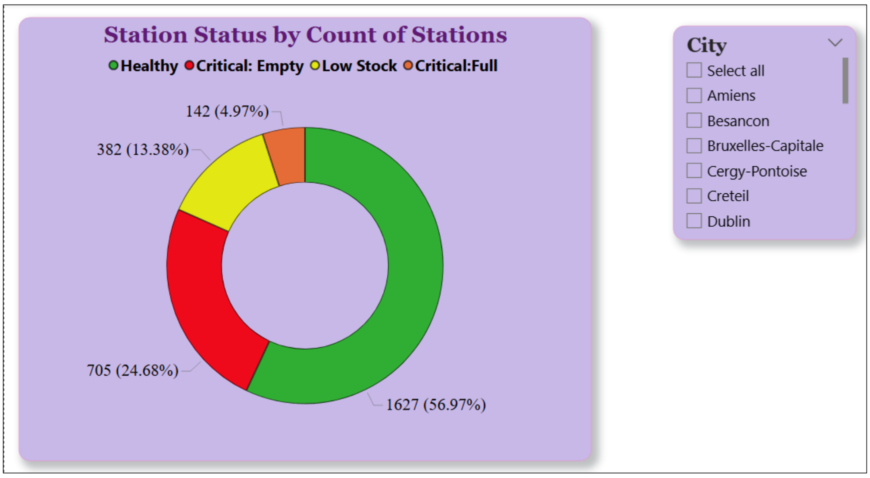
Geographic Distribution of Station Status:

- The key insight from the Geographic Distribution of Station Status map is that system imbalance is highly localized, with significant **"Critical: Empty" (red)** and **"Critical: Full" (orange)** clusters across Europe.
- This reveals that cities like Seville and regions in France face severe service gaps, requiring targeted bike redistribution to move the network back to a **"Healthy" green state.**



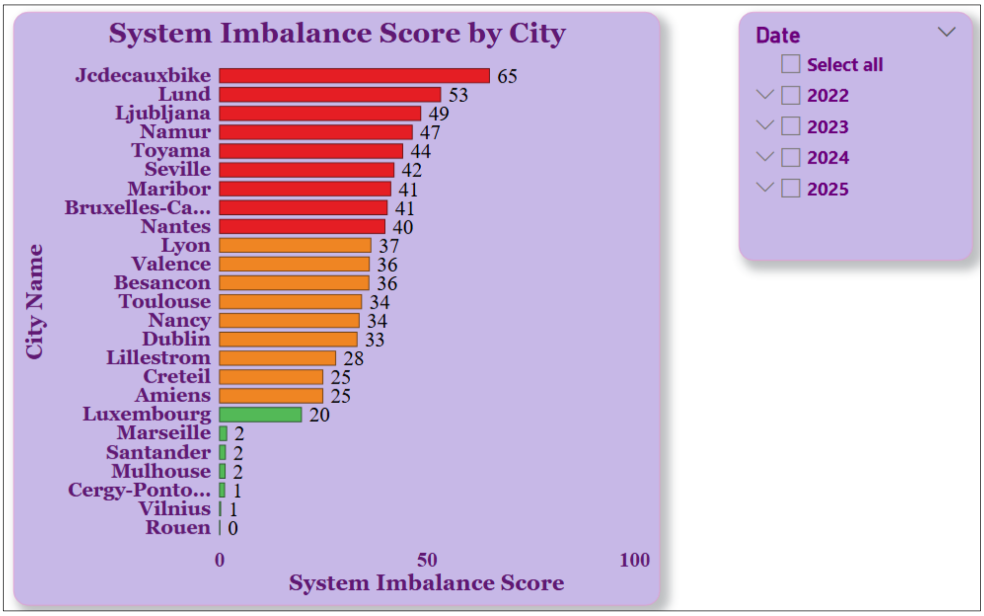
Station Status by Count of Stations:

- The Station Status by Count of Stations chart reveals that nearly 30% of the network is currently in a "Critical" state, with 24.68% (705 stations) being completely empty and unable to serve users.
- While over half the network remains "Healthy," the significant volume of empty and low-stock stations suggests a major need for bike redistribution to maintain service availability.



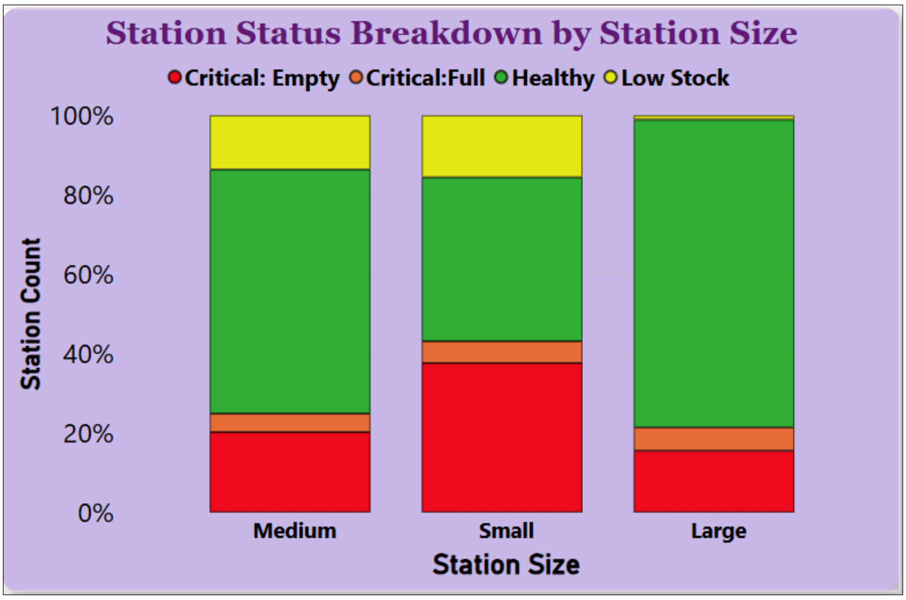
System Imbalance Score by City:

- ❖ The System Imbalance Score by City chart reveals a massive performance gap between contracts, with Jcdecauxbike (65) and Lund (53) being the most inefficient.
- ❖ In contrast, cities like Rouen, Vilnius, and Marseille maintain nearly perfect balance with scores between 0 and 2, setting the benchmark for operational excellence in the network.



Station Status Breakdown by Station Size:

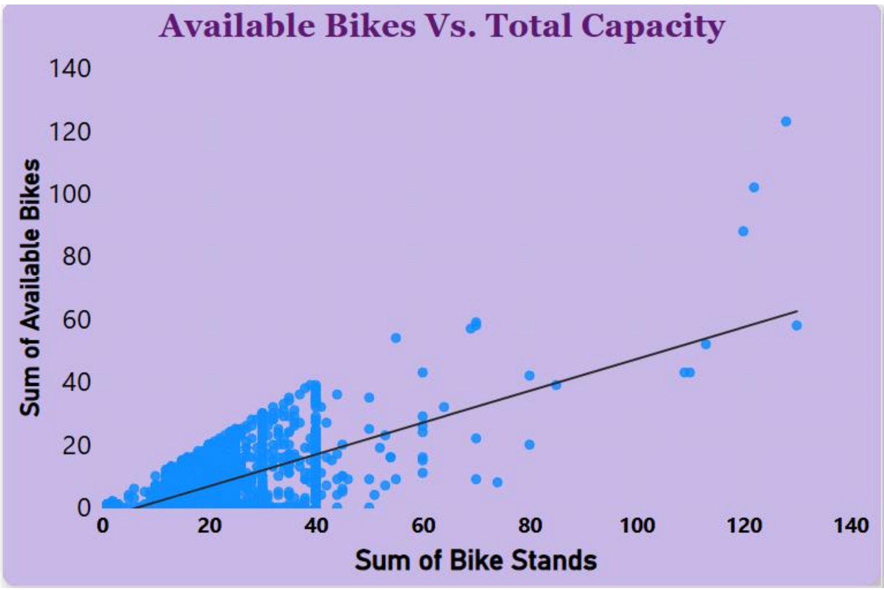
- ❖ The Station Status Breakdown by Station Size chart reveals that **Small** stations are the most vulnerable to inventory issues, with nearly 40% reaching a "Critical: Empty" state.
- ❖ In contrast, **Large** stations maintain the highest efficiency, with over 75% of stations remaining in a "Healthy" operational status.



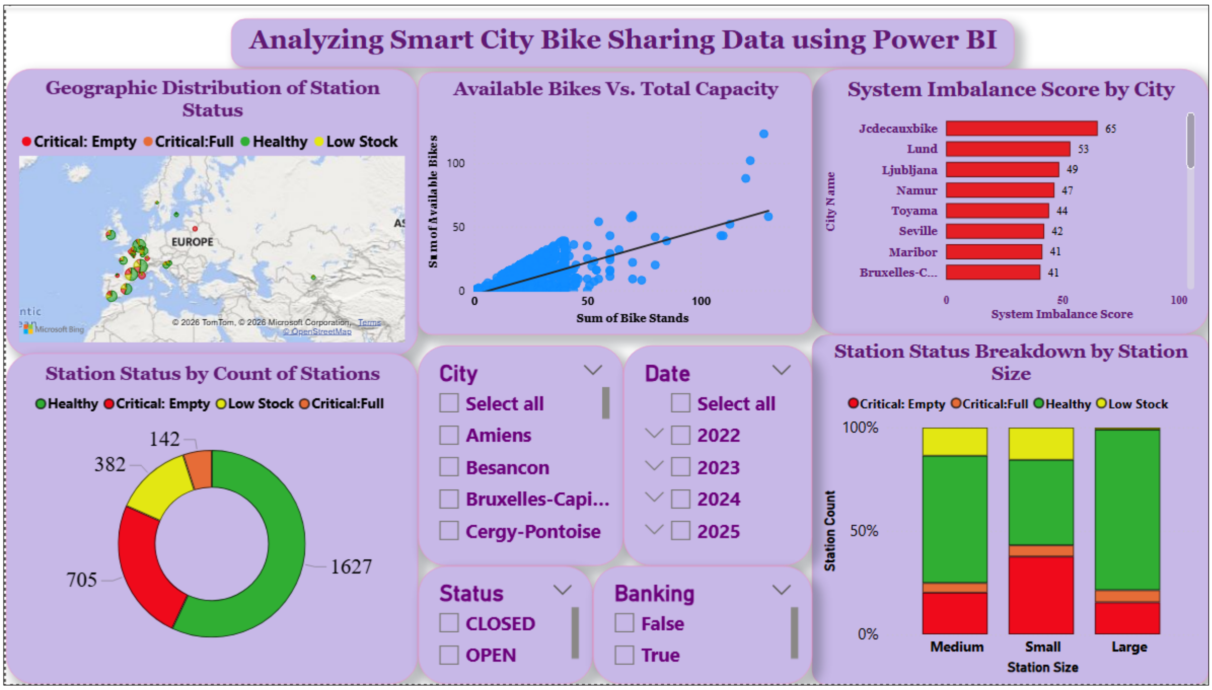
Available Bikes Vs. Total Capacity:

- ❖ **Identification of Underutilized Infrastructure:** Multiple data points fall significantly below the trend line, representing stations that are "Overbuilt" or "Critical: Empty". For instance, some stations with approximately 130 stands hold fewer than 60 bikes, suggesting wasted capacity.

- Conversely, several stations sit far above the trend line, such as a station with roughly **125 stands** holding nearly **125 bikes**. These represent "Critical: Full" risks where users may soon be unable to return bikes.



Report



9. Insights & Conclusions

Key Findings

- Approximately 30% of the global network is in a "Critical" state, with **24.68%** of stations being **completely empty**.

- There is a massive performance gap between cities; **Marseille** and **Rouen** maintain **near-perfect balance (Score 0–2)**, while **Jcdecauxbike** and **Lund** suffer from **extreme inefficiency (Score 53–65).**
- Small stations are the most fragile, with a 40% Critical: Empty rate, whereas Large stations are the most stable, with 75% remaining "Healthy".
- The scatter plot identifies significant "Overbuilt" outliers—stations with over 130 stands but fewer than 60 bikes available.

Analysis Insights

- Prioritize rebalancing efforts in cities with an Imbalance Score above 40 to move bikes from "Full" outliers to "Empty" clusters.
- For Small stations with high empty rates, consider physical expansion or implementing "Bonus" incentives to encourage users to return bikes to those specific locations.

10. Conclusions

The integration of Power BI proved effective for end-to-end data analysis, from raw data to visual reporting.